AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Problem Statement

a smart grid managing system that uses machine learning to effectively manage local renewable sources such as solar power, high demand zones (EV charging household) efficiently controlling bidirectional flow of electricity and manages overall transformer health.

Problem Understanding

- a wider range of people are being affected by this problem. this includes: civilians,discoms(tangedco) and renewable energy utilization
- it is caused by the increasing installation of solar power sources in housholds and extensive ev charging
- the stakeholders are Power distribution companies/discoms, Municipalities and smart city authorities, Ev owners and charging station operators, Rooftop solar consumers, State load dispatch centers.

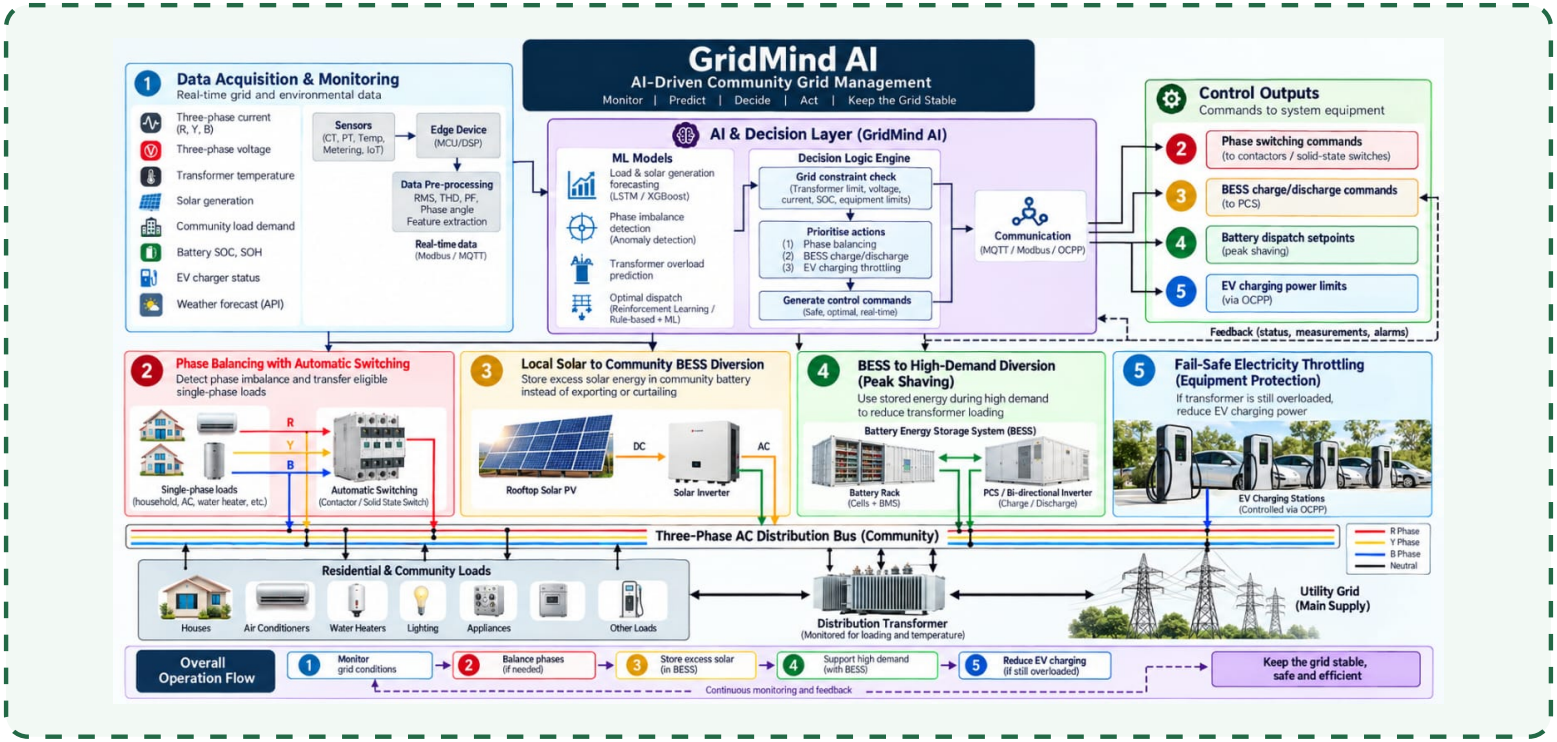
Existing Gaps & Challenges

- researchers suggest drawing power from charging evs to manage grid. but civilians wont prefer their ev's battery being charged and discharged repeatedly(this inturn degrades the battery) without actually using the vehicle.
- papers suggest using ai and ml models to manage the renewable sources for ev charging. but both peak at opposite times of the day(solar energy peaks at noon while civilians charge their ev mostly at evening)
- identification of solar source and ev draw as a major cause for grid instability and ultimately transformer breakdown is not much given attention.

Our Proposed Solution

- the core idea of the product is to moderate the health of the nearby transformer of imminent power surges caused by bidirectional flow of current(due to solar sources) and overheating(due to high power draw)
- at the higher level the system ensures that the transformer damage is reduced and power outages under sub-secondary grids
- as a result the civilians will get uninterrupted power supply and transformer health degradation will significantly reduce

Technical Approach



the system runs by using a set of sensors to record the reading and an ai to compare them in real time

Tech Stack

Frontend: Next.js Dashboard

Backend: Python

Database: InfluxDB

AI/ML, APIs, or Hardware: TensorFlow,Modbus RTU,ESP32-S3

Innovation & Uniqueness

- Grid Sense AI- takes a innovative approach by involving AI in the form of phase imbalance, overload prediction and recommending redistribution of loads it also helps in coordinating renewable energy and Battery Energy Storage System (BESS) technology during high demand in transformers due to EV charging
- Our approch combines phase imbalance,BESS with Grid Sense AI coordinates to store surplus renewable energy and discharge during peak hours.
- grid sense AI aims to combine ai based three phase imbalanced detection with BESS charging/discharging decesion in one coordinate system unlike system fouced mainly on forecasting or storage scheduling

Feasibility & Scalability

- **Feasibility:** grid sense AI can be developed using mathlab or python to simulate three phase loads , detect and perform phase imbalance using machine learning , and model a (BESS) .This system will recommend suitable load redistribution and battery charging or discharging actions
- **Scalability:** the device can be implemented near almost all of the transformers around the country due to its high flexibility in operation. device when installed, collects data which unique to that particular grid and derives solutions accordingly.
- **Risks:** since the data is being also remotely accessed through iot systems, potential cyber attacks can happen to the device seizing the device from operating properly.

Cost for Product

- Development or hardware cost: the hardware coponets including dedicated micro computers for running the machine learning program, switches and relays and senors all add upto around rupees 10000 per device.
- Hosting or maintenance cost:maintenance cost will low because the system is completely automated. only occational checks are required
- Estimated total:10000(product cost)+1500-2000 per year(maintenance cost)=34000 for the first year of installation followed by 20000 anual expenses.

Use Cases

- Use case 1: when there exists a overproduction of solar power, reverse passage of electricity happens in the grid. if the reverse electricity reaches transformer, it causes unwanted heating. the device stops the reverse voltage from traveling to the transformer and diverts to BESS system.
- Use case 2: at evening time when a lot of EVs are plug intoa grid, to much draw in current takes place which drops the voltage from 400v to 160 v. the device takes power from BESS and delivers it into the grid restoring operational voltage.
- Use case 3: in worst case scenarios if the device senses that the transformer overheats, the device performs thermal throttling to protect the transformer from getting damaged.

Impacts and Benefits

- this highly benifits discoms and common people. it helps discoms by reducing the transformer damage and burnout leading to replacement. it aids common people by reducing the number of operator induced blackouts to prevent components.
- the renewable energy can be used efficiently instead of going to waste and a lot less of burnt transformers are going to waste.
- less expense to replacing transformer and uninterrupted power supply for people.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
7	7.3	[affordable and clean energy]	GridSense AI directly supports energy efficiency by optimizing local distribution networks, preventing thermal wastage
9	9.1	Industry, Innovation, and Infrastructure	The project upgrades legacy electrical grids into resilient, smart infrastructure by deploying edge-AI hardware.

Individual Member Contribution

Member	Contribution or responsibility
Dhivya Balan	analusing the problem and collecting existing data
Krithik	statistically estimates the amount of work that the modern system should do to manage modern grid problems
Botla Anil	analyses the importance of using the ai to replace the work of humans in this specific scenario
Bhuvanesh	prototing and undestanding the working of ml model

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